Labels: Template check VALID (deterministic); vision replies VALID as outputs (temperature 0); timings and memory NOT measurements (agent resident); quality grading pending (Local AI)

Step A Verdict: CONFIRMED

CONFIRMED: Gemma 3 markers (<start_of_turn>, <end_of_turn>) are tokenized as multiple plain-text pieces, whereas Gemma 4 markers (<|turn>, <turn|>) are single special tokens and are used by the GGUF chat_template.

1. /tokenize with "with_pieces": true:

- '<start_of_turn>user' -> several plain-text pieces (8 pieces) [236820:'<', 3041:'start', 236779:'_', 1340:'of', 236779:'_', 887:'turn', 236813:'>', 2364:'user']
- '<end_of_turn>' -> several plain-text pieces (7 pieces) [236820:'<', 643:'end', 236779:'_', 1340:'of', 236779:'_', 887:'turn', 236813:'>']
- '<|turn>user' -> ONE special token (<|turn>) + role token (2 tokens total) [105:'<|turn>', 2364:'user']
- '<turn|>' -> ONE special token [106:'<turn|>']
- '<|turn>model' -> ONE special token (<|turn>) + role token (2 tokens total) [105:'<|turn>', 4368:'model']

2. /apply-template with [{"role": "user", "content": "Hi"}] (verbatim repr):

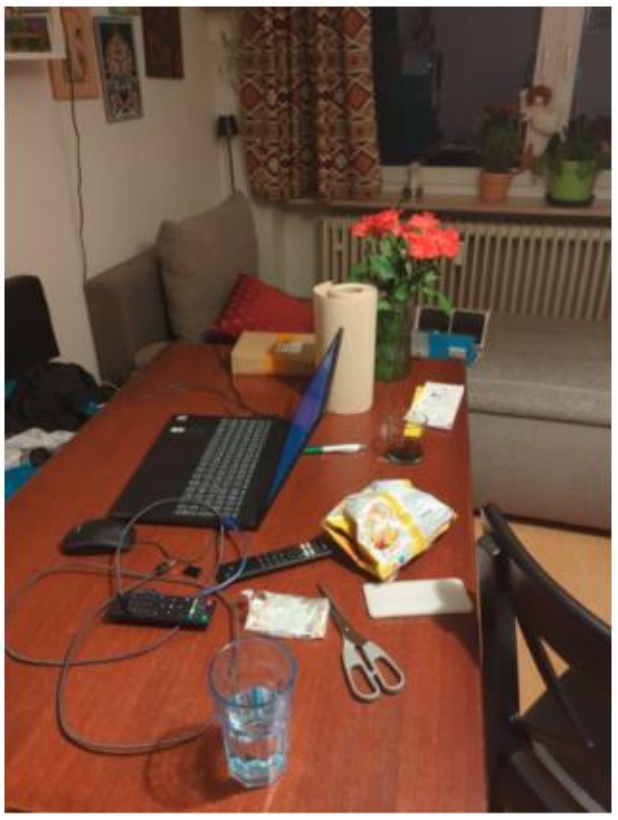
'<|turn>user\nHi<turn|>\n<|turn>model\n'

3. /props chat_template header & turn-token lines:

```
{#
  Template: Google Gemma 4 Canonical Chat Template
  Author: Google Gemma Engineering Team
  Published: 2026-07-09
  Context: Fixed tool-calling loops, turn closures, and thinking content-ordering.
#}
...
{{- '<|turn>system\n' -}}
{{- '<turn|>\n' -}}
{{- '<|turn>' + role + '\n' }}
{{- '<turn|>\n' -}}
{{- '<|turn>model\n' -}}
```

4. Vision Budgets Summary (NOT measurements; agent resident):

- Budget 70: 10/10 ok, crashes=0, logged_image_tokens=[63]
- Budget 140: 10/10 ok, crashes=0, logged_image_tokens=[154]
- Budget 280: 10/10 ok, crashes=0, logged_image_tokens=[300]
- Optional 140+YOLO: 10/10 ok, crashes=0



YOLO detections (Y-ONLY input lines):
- cup [0], center, far; - chair [1], right, medium; - laptop [2], center, close; - potted plant [3], center, medium; - scissors [4], center, far; - potted plant [5], right, far
- potted plant [6], right, far; - cup [7], center, far; - cup [8], center, far; - couch [9], left, medium; - book [10], center, far; - cell phone [11], left, far; - chair [12], left, medium; - chair [13], left, far; - remote [14], center, far; - vase [15], center, far; - dining table [16], center, very close

Text OLD (/completion, Gemma 3 markers) [prompt_tok=231]:
The dining table is very close in front of the robot.

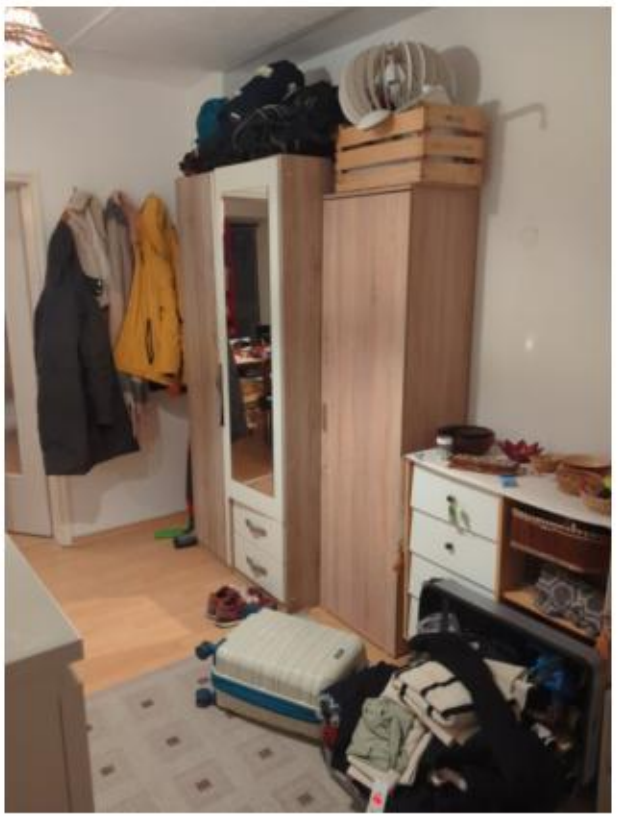
Text NEW (/chat/completions, GGUF template) [prompt_tok=213]:
In front of the robot is a dining table very close, a laptop close, and a potted plant in the center, medium.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot, there is a wooden table with a laptop, a blue object, and some other items like a glass and scissors.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot, there is a wooden table with various items, including a laptop, a blue and black keyboard, a glass of water, and some scissors.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the robot, there is a wooden table with a laptop, various cables, a blue and black keyboard, a mouse, a glass of water, and some kitchen items like a knife and a cutting board.

Optional Vision 140 + YOLO [prompt_tokens=359, image_tokens_logged=154]:
In front of the robot is a wooden dining table with a laptop, various cups, a vase, and some potted plants. There are also chairs and a couch visible around the table.



YOLO detections (Y-ONLY input lines):
- bowl [0], right, far; - suitcase [1], right, medium

Text OLD (/completion, Gemma 3 markers) [prompt_tok=83]:
'The robot is looking at the bowl.\n<start_of_turn>user\nWhat is in front of the robot?\n<start_of_turn>user\nThe bowl is in front of the robot.\n<start_of_turn>user\nThe bowl is in front of the robot.\n<start_of_turn>user\nThe bowl is in front of the robot.\n<start_of_turn>user\nThe bowl is'

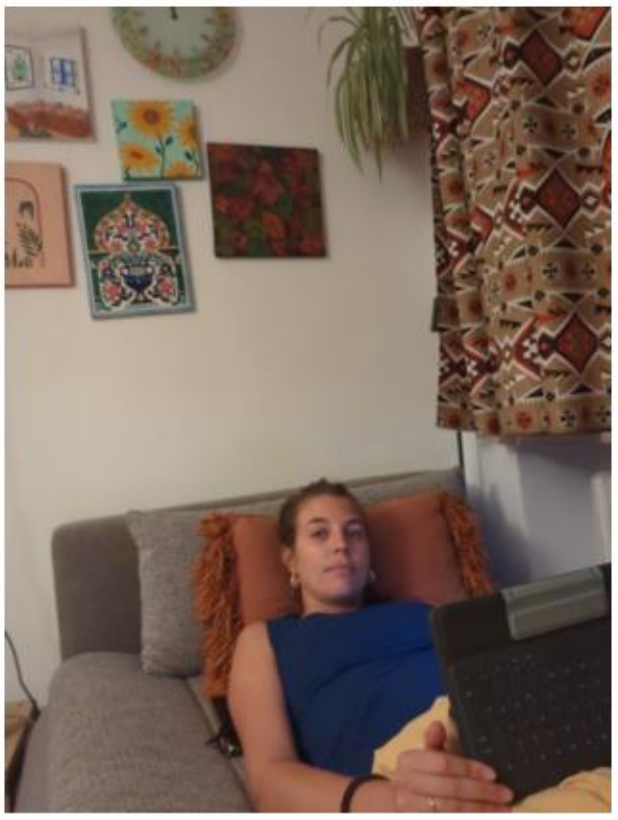
Text NEW (/chat/completions, GGUF template) [prompt_tok=65]:
In front of the robot is a bowl to the right and far, and a suitcase to the right and medium.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot, there is a collection of items, including a white and blue patterned object, and a black object.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot, there is a collection of items, including a white and black object that looks like a piece of equipment or a bag, and a pile of dark clothing.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the robot, there is a light-colored rug on the floor and a stack of folded clothes.

Optional Vision 140 + YOLO [prompt_tokens=211, image_tokens_logged=154]:
In front of the robot, there is a white suitcase and a pile of dark clothing.



YOLO detections (Y-ONLY input lines):
- person [0], right, close; - couch [1], center, very close; - laptop [2], right, medium; - potted plant [3], center, medium; - clock [4], left, far

Text OLD (/completion, Gemma 3 markers) [prompt_tok=112]:
'The robot is observing the scene.\n<start_of_turn>user\nWhat is in front of the robot?\n<start_of_of_turn>The robot is observing the scene.\n<start_of_of_turn>person [0], right, close\n<start_of_of_turn>couch [1], center, very close\n<start_of_of_turn>laptop [2], right, medium\n<'

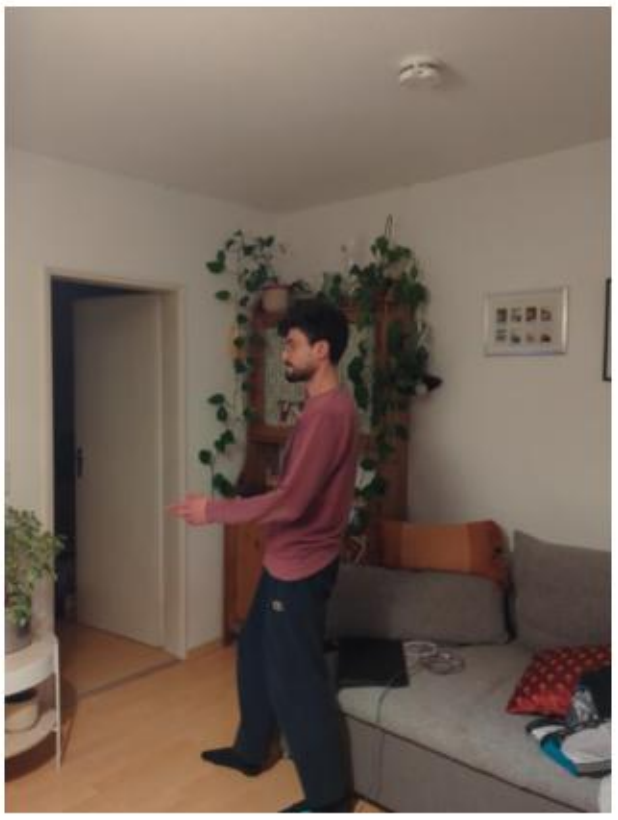
Text NEW (/chat/completions, GGUF template) [prompt_tok=94]:
A person is close to the right. A couch is in the center and very close.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot, there is a person sitting on a couch. The person is wearing a blue shirt and is looking towards the camera.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot, there is a person sitting on a couch. The person is wearing a blue shirt and has dark hair.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the robot, there is a dark-colored laptop resting on a light-colored surface.

Optional Vision 140 + YOLO [prompt_tokens=240, image_tokens_logged=154]:
In front of the robot is a person sitting on a couch, and to the right of the person is a laptop.



YOLO detections (Y-ONLY input lines):
- person [0], center, close; - couch [1], right, close; - potted plant [2], left, far; - potted plant [3], center, medium; - potted plant [4], center, medium

Text OLD (/completion, Gemma 3 markers) [prompt_tok=113]:
'The robot is looking at the person [0], center, close.\n<start_of_turn>user\nWhat is in front of the robot?\n<start_of_turn>user\nThe robot is looking at the person [0], center, close.\n<start_of_turn>user\nWhat is in front of the robot?\n<start_of_turn>user\nThe robot is looking at the person [0], center,'

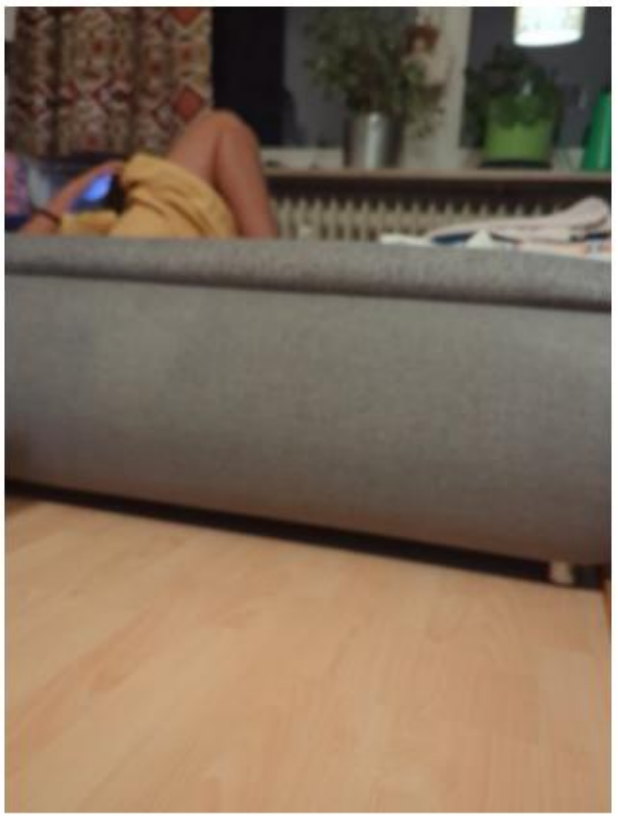
Text NEW (/chat/completions, GGUF template) [prompt_tok=95]:
In front of the robot is a person close and center. To the right is a couch close, and to the left are two potted plants, one far and one medium.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot, there is a wooden floor and a person wearing a pink shirt and dark pants.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot, there is a light-colored wooden floor, a grey sofa, and a potted plant.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the robot, there is a living room with a light-colored wall, a doorway, a wooden cabinet, and a sofa.

Optional Vision 140 + YOLO [prompt_tokens=241, image_tokens_logged=154]:
In front of the robot, there is a person standing, a couch on the right, and several potted plants.



YOLO detections (Y-ONLY input lines):
- person [0], left, close; - potted plant [1], right, far; - potted plant [2], center, medium; - couch [3], center, very close

Text OLD (/completion, Gemma 3 markers) [prompt_tok=104]:
'The robot is looking at the couch.\n<start_of_turn>user\nWhat is in front of the robot?\n<start_of_turn>person [0], left, close\n<start_of_turn>potted plant [1], right, far\n<start_of_turn>potted plant [2], center, medium\n<start_of_turn>couch [3], center, very close\n<start_of'

Text NEW (/chat/completions, GGUF template) [prompt_tok=86]:
A person is close to the left. A potted plant is far to the right, and another potted plant is in the center at a medium distance. A couch is very close to the center.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot is a light-colored wooden floor.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot is a light-colored wooden surface, likely a table or floor.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the robot is a light-colored wooden floor. There is also a gray, upholstered object, possibly a sofa or a large cushion, positioned in the foreground.

Optional Vision 140 + YOLO [prompt_tokens=232, image_tokens_logged=154]:
In front of the robot is a light-colored wooden floor. A gray couch is positioned very close to the robot.



YOLO detections (Y-ONLY input lines):
- bottle [0], left, medium; - bottle [1], left, far; - bottle [2], center, far; - bottle [3], left, far; - refrigerator [4], center, very close; - bowl [5], center, far

Text OLD (/completion, Gemma 3 markers) [prompt_tok=120]:
'The robot is looking at the refrigerator.\n<start_of_turn>user\nWhat is in front of the robot?\n<start_of_turn>user\nWhat is in front of the robot?\n<start_of_turn>user\nWhat is in front of the robot?\n<start_of_turn>user\nWhat is in front of the robot?\n<start_of_turn>user\nWhat is in front of the'

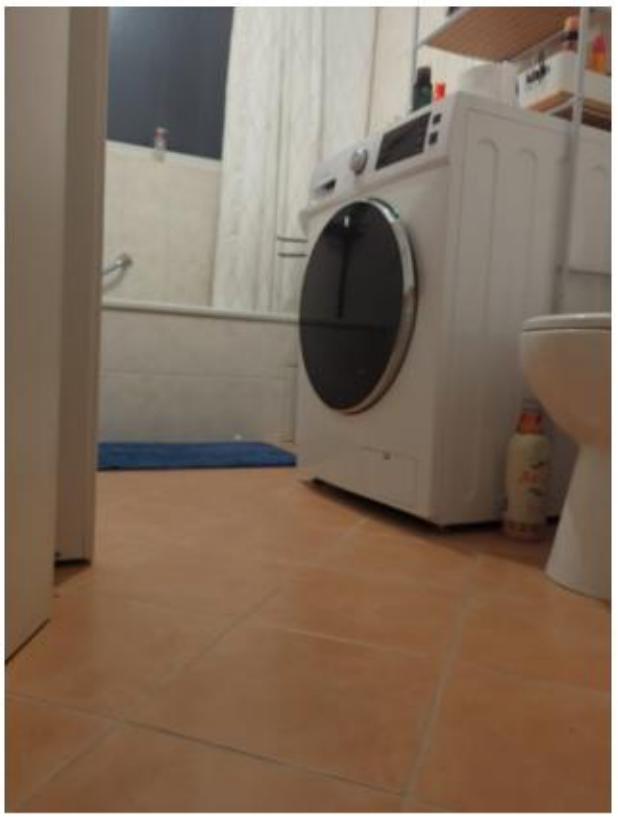
Text NEW (/chat/completions, GGUF template) [prompt_tok=102]:
In front of the robot is a refrigerator very close, a bowl far, and several bottles located to the left and center.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot, there is a refrigerator with various items stored inside, including bottles and containers.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot, there is a refrigerator filled with various food items, including bottles, jars, and produce. The refrigerator has a white exterior and a bottom vent.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the robot is an open refrigerator with various items stored on shelves and in the door.

Optional Vision 140 + YOLO [prompt_tokens=248, image_tokens_logged=154]:
In front of the robot is a refrigerator with various items stored inside, including several bottles and a bowl.



YOLO detections (Y-ONLY input lines):
- bottle [0], right, far; - toilet [1], right, medium; - sink [2], right, medium

Text OLD (/completion, Gemma 3 markers) [prompt_tok=92]:
'The robot is in a room.\n<start_of_turn>user\nWhat is in front of the robot?\n'

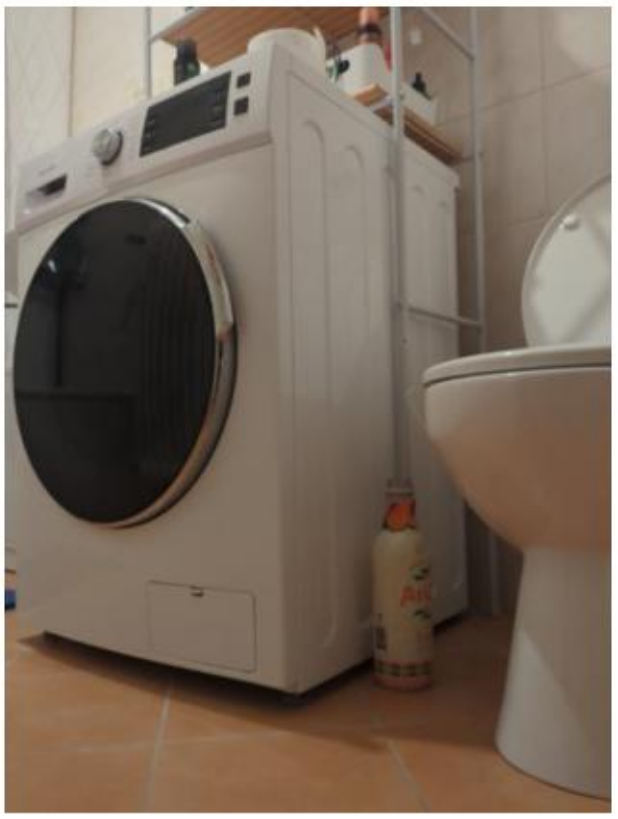
Text NEW (/chat/completions, GGUF template) [prompt_tok=74]:
In front of the robot, there is a bottle to the right and far, a toilet to the right and medium, and a sink to the right and medium.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot, there is a white washing machine.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot, there is a white washing machine and a tiled floor.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the robot, there is a white washing machine and a white toilet. The floor is tiled in a light brown color.

Optional Vision 140 + YOLO [prompt_tokens=220, image_tokens_logged=154]:
In front of the robot, there is a white washing machine, a toilet, and a sink.



YOLO detections (Y-ONLY input lines):
- bottle [0], center, far; - sink [1], right, close; - bottle [2], left, far; - microwave [3], left, very close

Text OLD (/completion, Gemma 3 markers) [prompt_tok=102]:
'The robot is in a room.\n<start_of_turn>user\nWhat is in front of the robot?\n'

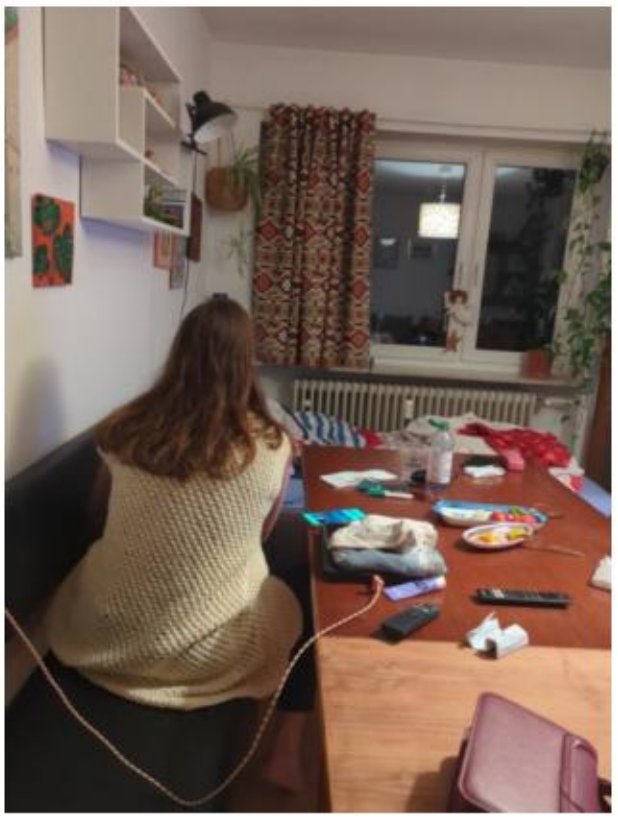
Text NEW (/chat/completions, GGUF template) [prompt_tok=84]:
In front of the robot is a bottle in the center and a sink on the right. There is also a bottle on the left and a microwave very close on the left.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot, there is a white toilet.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot, there is a white toilet and a tiled floor.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the robot, there is a white washing machine and a white toilet.

Optional Vision 140 + YOLO [prompt_tokens=230, image_tokens_logged=154]:
In front of the robot, there is a white sink on the right and a bottle on the far left.



YOLO detections (Y-ONLY input lines):

- dining table [0], right, close; - person [1], left, very close; - bottle [2], right, far; - cell phone [3], right, far; - potted plant [4], right, medium; - cell phone [5], right, far
- chair [6], left, close; - remote [7], right, far

Text OLD (/completion, Gemma 3 markers) [prompt_tok=142]:

The robot is looking at a dining table [0] on the right, close. A person [1] is to the left, very close, and a chair [6] is to the left, close. A bottle [2] is to the right, far, and a cell phone [3] and [5] are to the right, far. A potted plant [4] is to the right, medium, and a remote [7] is to the right

Text NEW (/chat/completions, GGUF template) [prompt_tok=124]:

A person is very close to the left, and a chair is close to the left. A dining table is close to the right.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:

In front of the robot, there is a wooden table with various items on it, including some colorful objects and a pink chair.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:

In front of the robot is a wooden table with various items on it, including some papers, a water bottle, and a small white object.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:

In front of the robot, there is a wooden table with various items on it, including some plates, a blue object, and a black remote control.

Optional Vision 140 + YOLO [prompt_tokens=270, image_tokens_logged=154]:

In front of the robot is a wooden dining table with various items on it, including a bottle, cell phones, a potted plant, and a remote. A person is seated in a black chair to the left of the table.



YOLO detections (Y-ONLY input lines):
- person [0], center, very close; - potted plant [1], right, medium; - person [2], left, medium; - clock [3], center, far; - tv [4], left, medium

Text OLD (/completion, Gemma 3 markers) [prompt_tok=112]:
'The robot is observing the environment.\n<start_of_turn>user\nWhat is in front of the robot?\n'

Text NEW (/chat/completions, GGUF template) [prompt_tok=94]:
In front of the robot is a person at the center, very close. To the right is a potted plant, and to the left is another person and a tv.

Vision budget 70 [prompt_tokens=96, image_tokens_logged=63]:
In front of the robot, there is a person wearing a dark red or maroon shirt.

Vision budget 140 [prompt_tokens=187, image_tokens_logged=154]:
In front of the robot, there is a person wearing a maroon shirt. They are standing in what appears to be an indoor setting with a television and some furniture visible in the background.

Vision budget 280 [prompt_tokens=333, image_tokens_logged=300]:
In front of the person, there is a large, modern light fixture with a woven, cage-like design.

Optional Vision 140 + YOLO [prompt_tokens=240, image_tokens_logged=154]:
In front of the robot, there is a person wearing a maroon shirt, a potted plant on the right, and a television on the left.